EPIDURAL GUIDELINES

University Teaching Hospital

Contents

Epidural Guidelines **2**
Aims and Standards for Epidural Analgesia in Labour 2
Indications 2
Contraindications 2
Absolute 2
Relative 2
Recommended Procedure 3
Fetal monitoring 3
Consent 3
Techniques 3
Documentation 4
Monitoring 4
Indications to Call Anaesthetist 4
Removal of Epidural Catheter 4
Weight >60 kg and blood loss <1500 mL 4
Weight <60 kg and PPH*, or Blood loss >1500 mL, or Risk of ongoing bleeding (e.g., sepsis, thrombocytopenia, abruption or other consumptive process) 5
Recovery from Epidural 5
References 5
Drug Preparation/Administration **6**
Standard Mix Preparation 6
Drug storage 6
Drug Administration for Labour analgesia 6
1. TEST DOSE - This is the initial dose of medication given down the epidural upon first siting it. Its purpose is to reveal a catheter inadvertently placed in the intrathecal or intravascular space 6
2. LOADING DOSE - This is the additional dose that follows a negative test dose. Its purpose is to achieve appropriate spread of local anaesthetic in the epidural space 6
3. CONTINUOUS INFUSION - Once successfully loaded, epidural analgesia is maintained by continuous infusion of standard mix via programmed syringe driver 6
4. CLINICIAN BOLUS/"TOP UP" - This is an additional bolus to troubleshoot an inadequate epidural 7
Epidural Top-Up For Theatre **8**
Assessment 8
Top-up 8
Rationale for choice of drug 8
Drug Preparation 8
Drug Administration 8
References 9

Mobilising with Epidurals 10
Aim 10
Contraindications to mobilising with an epidural 10
Extent of Mobilisation 10
Monitoring and ensuring a safe environment for mobilisation 10
Mobilisation assessment 10
Intravenous Fluid Administration and Blood Pressure Monitoring 13
Intravenous Cannulation 13
Blood Pressure Monitoring Following an Epidural 13
Management of Hypotension 13
Blood Pressure Maintenance with Phenylephrine Infusions 14
Introduction 14
References 14
Epidural Troubleshooting 16
Causes of Ineffective Epidurals 16
Subcutaneous 16
Subdural 16
Subarachnoid 16
Intravascular 17
Unilateral Block 17
Insufficient Activation 17
Catheter Migration 17
Abnormal Anatomy 17
Sacral Sparing 17
Precipitant Labour 17
Patient Expectations 17
Management of Insertion and Placement Difficulties 17
Inability to Thread the Catheter 17
Blood in the Catheter 17
Paraesthesia 17
Management of Ineffective Epidurals 18
Assessment 18
No block; low block 18
Poor sacral coverage 18
Unilateral block 18
Missed segment 18
Back pain/rectal pressure 18
Standard mix or 0.25% Levobupivacaine? 19
Adjuvant Drugs 19
Fentanyl 19
Re-siting 19
References 19
Treatment of Post-Dural Puncture Headache 21
Management 21
Conservative management 21
Pharmacological management 21

Caffeine..... 21
Other medication..... 21
Epidural blood patch..... 21
Epidural blood patch technique..... 21
References..... 22
High Regional or Total Spinal Block..... 23
Management..... 23
Equipment immediately required..... 23
Management technique..... 23
Post-event..... 23
References..... 23

Epidural Guidelines

Aims and Standards for Epidural Analgesia in Labour

Establish effective analgesia before a mother has severe uncontrolled pain
Attend within 60 minutes of request for analgesia, subject to staff availability
Maintain analgesia during the second stage of labour
Convert analgesia to anaesthesia for instrumental or operative delivery when required (anaesthetist to topup)

Indications

Maternal request
Prolonged and painful labour
Malpresentation
Anticipated or actual instrumental delivery
Trial of labour (in anticipated surgical delivery)
Pre-eclampsia
Diabetes
Obesity (BMI > 35)
Uncoordinated uterine activity managed by oxytocin infusion
Multiple pregnancy
Cardiac and respiratory disease

Contraindications

Absolute

Lack of appropriately trained staff (includes a trained midwife for monitoring)
Refusal by mother
Full anticoagulant therapy
Coagulopathy
Hypovolaemia
Local anaesthetic allergy

Relative

Local or systemic sepsis (mild pyrexia is not a contraindication)
Raised intracranial pressure
Fixed cardiac output

Recommended Procedure

Establish the indication and review obstetric, medical and anaesthetic history.
Full anaesthetic pre-operative assessment

Fetal/maternal monitoring

CTG for a period of 30 minutes before an epidural is sited
CTG should be interpreted before starting the procedure
Fetal heart is monitored by continuous CTG whilst epidural infusion running.
Full set of baseline maternal observations should be taken and documented prior to procedure.

Consent

Provide mother with the laminated Epidural Information Card if she is considering an epidural and before she is distressed with pain.
Obtain an informed, verbal consent, which should include a description of the technique, the benefits and an explanation of possible complications including:

Backache	Short term only (no long term risk)
Itch (short term)	1 in 3
Failure/inadequate block	1 in 10
Hypotension	1 in 20
Nausea and vomiting	1 in 20
Post dural puncture headache	1 in 100
Temporary nerve damage	1 in 1,000
Permanent nerve damage	1 in 10,000
Meningitis	1 in 50,000
Epidural haematoma	1 in 100,000
Epidural abscess	1 in 100,000
Surgical/instrumental delivery	Unknown but increased risk

Technique

- Ensure resuscitation equipment available
- Intravenous access (16G or 18G cannula) obtained, and flushed with 0.9% saline
- Full asepsis (hat, mask, gown and gloves). Maintain sterile field
- Check your equipment. Prime filter and epidural catheter with saline to check patency.
- Use fresh sterile saline for each patient
- Clean back using iodine solution followed by methylated spirit. Allow to fully dry prior to needling.
- Clean lidocaine bottles with spirit prior to drawing (if already opened)
- Epidural insertion
    - Infiltration of 2% lidocaine to skin and sub-cut tissues
    - Use Tuohy needle to locate epidural space using a **continuous pressure, loss of resistance to saline** technique
    - Make note of the depth of the needle at loss or resistance.

    - Thread epidural catheter through the Tuohy needle and remove the Tuohy.
    - Withdraw the catheter to a suitable length to leave 4-5cm in the epidural space. **Never withdraw the catheter through the Tuohy as this risks severing the catheter.**
- Confirm positioning with "**falling meniscus"** and "**negative aspiration"** test
    - Initial aspiration test should *not* be via the filter (very small risk of air lock giving false negative)
    - Aspiration should be negative both for blood and CSF
- Make sure epidural filter is connected firmly to epidural catheter by 'tug-test'
- Secure epidural catheter with a clean, dry, sterile dressing over site of entry. Include a loop of catheter to reduce risk of accidental catheter removal.
- Further secure with non-sterile strapping
- Give a "Test Dose" (see next section)
- If no adverse effects (see next section) after 5 minutes give a "Loading Dose" and start continuous infusion.
- Seek assistance if difficulty with insertion: e.g., senior help, ultrasound scan, move to theatre.
- In some instances, commencing labour analgesia with a CSE may be indicated.

Documentation

A specific "UTH Epidural Chart" should be completed for each epidural.
This includes full preassessment, documentation of verbal consent, insertion documentation and prescription of epidural infusions (All to be completed by anaesthetist)
The attached monitoring chart should be completed by midwives.
Additional medications, fluids and additional notes should be documented in the patient notes as per standard.
All epidural patients should be entered into the epidural register - this can be found in the anaesthetic room in C-Block.

Monitoring

Pulse and blood pressure
- Every 5 minutes for 20 minutes after loading dose or clinician bolus, then every 30 minutes thereafter.

Pain assessment
- Pain scores to be documented hourly on epidural care chart

Motor block assessment
- Bromage score to be documented bilaterally hourly on epidural care chart

Pressure care
- Ensure one-hourly position change for care of pressure areas

Warn mother about risk of pressure sores

Indications to Call Anaesthetist

Systolic blood pressure decreases 20% below baseline
Patient feels nauseated or light-headed
Sensory level of block extends above the xiphisternum
Patient is unable to bend her knees or leg weakness is getting worse
Respiratory rate below 12 per minute
Analgesia inadequate
Instrumental/surgical delivery required
Disconnected filter
- Immediate discovery of disconnection: cut 10 cm off catheter with sterile scissors, wipe with alcohol/chlorhexidine skin wipe, allow to dry and reattach.
- Unknown duration of disconnection: re-site epidural.

Removal of Epidural Catheter

Under *no* circumstances should a woman ever go to the postnatal ward with an epidural catheter in-situ.

To remove the catheter, first remove all dressings/strapping. Gently pull the catheter which should easily come out. Ensure that the black/blue tip of the catheter is seen. Apply a sterile dressing to the site which can be removed after 1 hour.

Weight >60 kg and blood loss <1500 mL

Epidural catheter can be removed without needing to assess FBC or coagulation.
Low molecular weight heparin (LMWH) should be prescribed for at least 4 hours after the time of catheter removal, providing obstetric bleeding has stopped.

Weight <60 kg and PPH, or Blood loss >1500 mL, or Risk of ongoing bleeding (e.g., sepsis, thrombocytopenia, abruption or other consumptive process)*

Discuss the management of the epidural catheter with an anaesthetist prior to removal and LMWH prescription. FBC and coagulation may aid decision making.

*the volume of PPH of significance will depend on the individual patient if her weight is <60 kg. Remember that circulating volume in parturient at term is approximately 100 mL/kg; therefore if weight is 50 kg then 1250 mL = 25% of total blood volume.

Recovery from Epidural

During the recovery phase after an epidural, straight-leg raising should be used as a screening method to assess motor block. If the woman is unable to straight-leg raise 4 hours after epidural removal the anaesthetist should be called to assess whether the woman's care should be escalated to investigate the possibility of reversible causes of neurological injury.
Women should be informed of the likely timescale for resolution of their neuraxial block and encouraged to alert staff should this be delayed.

Follow up

All epidural patients should be followed up at least 24 hours post insertion of epidural. Follow up audit proforma is available in the department and will gather feedback on patient experience, epidural complications and analgesia.

References

Faculty of Pain Medicine of the Royal College of Anaesthetists. 2010. *Best practice in the management of epidural analgesia in the hospital setting*. Available at http://www.aagbi.org/sites/default/files/epidural_analgesia_2011.pdf (accessed 31 May 2016).

Safety guideline: neurological monitoring associated with obstetric neuraxial block 2020. A joint guideline by the Association of Anaesthetists and the Obstetric Anaesthetists' Association. 2020. https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.14993 (accessed 27 June 2022).

Massoth C, Wenk M. Epidural test dose in obstetric patients: should we still use it? 2019. Curr Opin Anaesthesiol; 32(3):263-267.

Gardner IC, Kinsella SM. 2005. Obstetric epidural test doses: a survey of UK practice. Int J Obstet Anesth;14(2):96-103.

Drug Preparation/Administration

Standard Mix Preparation

- Make up 2 x 50ml syringes of "Standard Mix" epidural infusion.
- Each 50ml syringe should include
    - 10ml of 0.5% bupivacaine (i.e. 50mg)
    - 1ml of fentanyl 50mcg/ml (i.e. 50mcg)
    - 39ml sterile (freshly opened) normal (0.9%) saline
- Result will be a total of 100ml of 0.1% bupivacaine and 1mcg/ml fentanyl split into 2 x 50ml syringes
- A fresh saline bottle must be opened for each epidural case. The same bottle may be used for saline for the procedure and saline for drug dilution. If appropriate the remaining saline can be given IV to the patient.

Drug storage

- One syringe will be used immediately for the patient including for test dose, loading dose and continuous infusion
- Second syringe must be clearly labeled with
    - **Drug name and concentration** (0.1% bupivacaine and 1mcg/ml Fentanyl in Normal Saline)
    - **Time and date drawn up**
    - **Name of anaesthetist drawing up**
    - **Time and date of expiry - *24 hours post drawing up***
- This syringe can be stored sealed, in the drug fridge, inside its packaging for a maximum of 24 hours before being disposed of. It can be used for the same patient or any other within its expiry time.

Drug Administration for Labour analgesia

In standard practice, the only medication to be delivered down the epidural in labour ward will be the "Standard Mix" as described above. All medications are given via the epidural filter. There are 4 principle dosing regimes to consider:

1. **TEST DOSE -** This is the initial dose of medication given down the epidural upon first siting it. Its purpose is to reveal a catheter inadvertently placed in the intrathecal or intravascular space.
    - Test dose should consist of 10mls of "Standard Mix". This can be administered manually via a syringe or as a programmed bolus via the syringe driver.
    - A negative test will result in no physiological changes at 5 minutes. Specifically no motor or perceivable sensory block, no haemodynamic changes and no signs/symptoms of systemic local anaesthetic dosing (tinnitus, periorbital paresthesia etc)
    - In the event of a positive test, the bolus should be immediately stopped. Any adverse effects should be managed as required. When safe to do so, the epidural catheter should be removed and re-siting considered.

2. **LOADING DOSE -** This is the additional dose that follows a negative test dose. Its purpose is to achieve appropriate spread of local anaesthetic in the epidural space.
    - Loading dose should consist of a further 5 to 10mls of standard mix. Lower doses considered for women of shorter stature and higher doses for those of taller stature.

3. **CONTINUOUS INFUSION -** Once successfully loaded, epidural analgesia is maintained by continuous infusion of standard mix via programmed syringe driver.
    - *Starting rate and rate adjustments are to be prescribed by anaesthetists only*
    - Suggested starting rate for continuous infusion is 10ml/hr
    - Variations on this can be considered for women of particularly short or tall stature
    - Infusion rate can be further adjusted based on inadequate or excessive block
    - It is recommended that rate changes occur in 2ml/hr increments/decrease

- *Remember to consider extra-epidural catheter as a cause of inadequate block*

4. **CLINICIAN BOLUS/"TOP UP" -** This is an additional bolus to troubleshoot an inadequate epidural
    - *Top-ups are to be prescribed and delivered by anaesthetists only*
    - In most circumstances, top-ups will consist of additional standard mix
    - Top-ups should ideally be delivered as a programmed bolus via syringe pump
    - In most circumstances, top-up should not exceed an additional 10mls of standard mix
    - *Following senior discussion*, *and under direct supervision of an anaesthetist*, additional preservative free opioid or stronger concentration local anaesthetic may be considered for epidural bolus.

Epidural Top-Up For Theatre

In the event that a patient with a functioning epidural needs a surgical intervention, the epidural can be "Topped-up" to achieve surgical anaesthesia.

Assessment

The epidural should first be assessed to ensure that it is reliable:

1) Discuss with the mother and midwife to ensure it has been working as expected
2) Review documentation
3) Check sensory block to cold or pin-prick sensation

If the block level is not equal on both sides, or the quality of the block is patchy, top-up is less likely to be successful. If there is significant doubt about the efficacy of the epidural, consider an alternative form of anesthesia e.g. spinal, GA or re-siting an new epidural if indicated.

Top-up

Up to 20mls of additional epidural medication can be introduced to achieve an adequate surgical block. The choice of medication depends on the urgency of the procedure.

1) If rapid conversion to anaesthesia is needed for an emergency procedure 2% Lidocaine with adrenaline should be used. See below for constitution guidance.
2) If non-urgent intervention is indicated 0.25% bupivacaine should be used.

Rationale for choice of drug

Whilst 2% Lidocaine with adrenaline provides a more rapid onset of block and potentially a more effective block, it involves higher total doses of local anaesthetic and encroaches on the maximum safe allowable dose for the average patient. Using 0.25% bupivacaine will take longer to achieve block and is more likely to require supplementary analgesics intra-op, however is safer from the point of view of Local Anaesthetic Systemic Toxicity (LAST) risk.

At the time of writing this document, UTH does not have access to intralipid for the management of LAST. As such the above recommendations have been considered to reduce the risk of LAST in no emergent cases whilst balancing this with the risk of harm to mother and baby in emergent cases.

Drug Preparation

As with preparation of "Standard Mix", freshly opened normal saline should be used for drug preparation to minimise infection risk.

1) 2% Lidocaine with 1:200,000 adrenaline -
    - Draw up 20ml 2% lidocaine into a 20ml syringe
    - In a separate 10ml syringe, dilute 1mg of adrenaline into 9ml of normal saline to give a concentration of 100mcg/ml adrenaline
    - Add 100mcg (1ml) of your diluted adrenaline to your 2% Lidocaine
    - Result it 2% Lidocaine with 5mcg/ml adrenaline (1:200,000). Technically this is 1.9% lidocaine and 4.76mcg/ml adrenaline as total solution is 21ml, however the drug mass is unaltered.
2) 0.25% bupivacaine
    - Draw 10ml of 0.5% bupivacaine into a 20ml syringe
    - Dilute to total 20ml with a further 10ml normal saline to make 0.25% bupivacaine.

Label epidural medications clearly and keep separate from medications intended for IV delivery.

Meticulous care must be taken not to give medications by the wrong route. Local anaesthetics intended for epidural use can cause LAST if given intravenously. Similarly, IV medications can cause significant harm if given via the epidural route - in particular, tranexamic acid (TXA) is highly neurotoxic.

Drug Administration

Ensure full monitoring and working IV access before commencing epidural top-up.
Top-up should be administered by a trained anaesthetist only and under direct supervision.
Ideally the full top-up should be given in the operating theatre. In emergency situations, top-up can be started in the labour ward, however the anaesthetist should then remain with the patient.
A maximum dose of 20ml of either solution should be used. It is recommended that this is given in 5ml aliquots with full monitoring and sensory block checked between each aliquot.
Time to onset of block will be quicker with Lidocaine (10 - 15 mins approx) when compared to bupivacaine (15 - 20 mins approx)
Titrate dose to achieve a sensory block of T4 bilaterally. This will usually require 15-20mls.

References

1. ***Conversion of labour epidural analgesia to surgical anaesthesia for emergency intrapartum Caesarean section***
   Desai, N. et al.
   BJA Education, Volume 20, Issue 1, 26 - 31
2. ***Hillyard SG, Bate TE, Corcoran TB, Paech MJ, O'Sullivan G. Extending epidural analgesia for emergency Caesarean section: a meta-analysis. Br J Anaesth. 2011 Nov;107(5):668-78. doi: 10.1093/bja/aer300. PMID: 21997149.***

Mobilising with Epidurals

Aim

The ideal epidural will provide adequate analgesia while still permitting full motor function. It is yet to be to what extent mobilisation will be feasible with the UTH standard epidural. Women with epidural analgesia may be able to mobilise off the bed during labour if it is deemed safe.
The decision to mobilise is dependent on:

- No contraindications to mobilising (see below)
- The woman being cardiovascularly stable
- The ability to provide continuous fetal monitoring
- The woman passing a mobilisation assessment

Note that mobility is not guaranteed and that motor block assessment must be carried out prior to any mobilisation. No women should mobilise if the midwife responsible for her care has concerns about safety.

Contraindications to mobilising with an epidural

Absence of a person to assist with mobilisation
Morphine/Pethidine in previous 4 hours
Any suspicion of drug or alcohol usage
Risk of cord prolapse
Postural hypotension or feeling faint on standing
Failure of motor power assessment (see below)
Woman not wishing to mobilise
If the midwife responsible for her care has concerns about safety

Extent of Mobilisation

Changing position in the bed, e.g., kneeling, on all fours
Transfer from bed to chair/commode
Walking around birthing room

Monitoring and ensuring a safe environment for mobilisation

Constant presence of one person to assist with balance at all times
Ensure women's observations and CTG stable for at least 15 minutes prior to mobilisation
CTG to be monitored continuously whenever possible. Any necessary disconnections to allow for movement between rooms etc should be minimised.
Mobilisation only to take place after woman has passed mobilisation assessment, carried out by the midwife.
Check for postural hypotension: blood pressure is checked immediately after standing and compared to the blood pressure when lying down. Postural hypotension is defined as a decrease in systolic blood pressure of 20mmHg or more when standing
Mobility assessment should be repeated at least hourly

Mobilisation assessment

Mobilisation assessment will be carried out by the midwife. Motor power assessment will be a 4-stage test. Parturient must be able to do all 4 stages below prior to walking:

Stage 1	Able to fully straight leg raise
Stage 2	Aware of normal sensation in soles of feet
Stage 3	Able to stand beside bed with two assistants for balance – not for support
Stage 4	Able to do three knee bends with help to balance only

Intravenous Fluid Administration and Blood Pressure Monitoring

Intravenous Cannulation

A 16G intravenous cannula should be inserted in the hand or arm before inserting an epidural. The cannula should be flushed with 0.9% saline to ensure patency.

Intravenous fluids (normally Ringer's lactate) are administered during labour for a number of specific purposes, for example:

- The infusion of drugs: e.g., oxytocin, antibiotics
- For dehydration following prolonged fasting, vomiting or diarrhoea
- To treat hypotension induced by an epidural or spinal anaesthetic

Blood Pressure Monitoring Following an Epidural

The midwife should check the blood pressure and fetal heart rate every 5 minutes for the first 20 minutes after the first epidural dose or after a top up

After 5 minutes in the prescribed position the mother may lie in whichever position she prefers subject to the fetal heart rate and blood pressure remaining stable

Management of Hypotension

If at any time the systolic blood pressure falls below 100 mmHg or by 20% of baseline the midwife should inform the anaesthetist and proceed as follows:

- Ensure the mother is in the lateral position (left or right depending on fetal heart)
- Elevate the feet
- Connect a 500 mL intravenous bag of Ringer's Lactate/Normal Saline solution and infuse it rapidly
- Give oxygen 4 L/min by Hudson mask
- If required, the anaesthetist will give intravenous phenylephrine or ephedrine
- Check the blood pressure and fetal heart rate every 5 minutes for the next 15 minutes or every minute if the blood pressure is falling.

Ephedrine must be available in the epidural room or labour ward at all times.

Blood Pressure Maintenance with Phenylephrine Infusions

Introduction

Phenylephrine, a direct acting α-agonist used for the treatment of hypotension, has been shown to be a useful vasopressor in obstetric anaesthesia and does not cause a decrease in fetal cord pH. As it has a relatively short half-life, it lends itself to administration by continuous IV infusion when treating hypotension induced by regional or general anaesthesia for operative / assisted delivery.

Indications	Contraindications
Hypotension following regional anaesthesia for assisted or operative delivery Hypotension following general anaesthesia for assisted or operative delivery	Known sensitivity to phenylephrine Hypertension Severe pre-eclampsia or eclampsia (relative)

Preparation	Regime
Inject 10 mg of phenylephrine into a 500 mL bag of 0.9% saline solution (final concentration 20 µg/mL) Draw up 50 – 60 mL into a syringe fitting into a syringe driver Connect the syringe to an anti-syphon extension set and flush through Load the syringe into the syringe driver and connect the extension tubing to the patient's IV access via a Coventry connector Use the *purge* or *bolus* function on the driver to ensure the extension is flushed prior to connecting Always administer together with a running IV infusion	*Operative or assisted delivery under spinal anaesthesia* Start infusion at a rate of 60 to 90 mL/hour immediately after injecting spinal drugs. Adjust infusion rate according to clinical response. *Operative or assisted delivery under top-up epidural anaesthesia or general anaesthesia* Start infusion at a rate of 30 to 90 mL/hour if clinically required. If phenylephrine is required consider giving a bolus of 40 µg (2 mL) before infusing. Adjust infusion rate according to clinical response. Use pump to administer 20 or 40 µg boluses as indicated Terminate or reduce infusions rapidly after delivery of the fetus if mother remains symptom-free

This guideline does not cover the use of phenylephrine infusions in any cases other than uncomplicated deliveries.
In the event of phenylephrine being unavailable, adrenaline infusion can be used?

References

Ngan Kee, W.D., Khaw, K.S., Ng, F.F. 2004. Comparison of phenylephrine infusion regimens for maintaining

maternal blood pressure during spinal anaesthesia for caesarean section. *British Journal of Anaesthesia*, 92(4): 469-74.

Epidural Troubleshooting

Epidural failure is a fact of life in obstetric anaesthesia. Its incidence is approximately 10 %. This chapter addresses causes for failure of epidural analgesia and the management of an ineffective epidural.

Summary of Causes of Ineffective Epidurals

Incorrect Placement	Correct Placement
Subcutaneous insertion Subdural Intrathecal Intravenous	Insufficient activation Catheter migration Abnormal anatomy Sacral sparing Precipitous labour Unrealistic expectations

Signs and Symptoms of Incorrectly Placed Epidural

Epidural Location	Block	Symptoms & Complications
Subcutaneous	No block	No analgesia Local tissue swelling
Subdural	Patchy sensory block + minimal motor block	High sensory block (including cervical) can occur
Intrathecal	Dose dependent; may have total spinal block	Haemodynamic and respiratory compromise Fetal bradycardia Post dural puncture headache
Intravascular	No block	Local anaesthetic toxicity Haemodynamic compromise/collapse

Causes of Ineffective Epidurals

Subcutaneous

This is the most common reason for failure to establish a block due to incorrect needle placement and is often the result of a false loss of resistance (loss of resistance occurring before entering the epidural space). It is more common in obese parturients and may be the result of the needle being off-midline or too shallow.

Subdural

The subdural space is a potential space between the dura mater and the pia/arachnoid mater. Since the arachnoid is not punctured there will be no CSF. Symptoms are variable but may be seen as a relatively rapid (5 – 15 minutes) onset of patchy sensory block with limited motor block. Subdural placement should also be suspected if there is a pronounced fall in blood pressure and/or heart rate in response to a test dose.

Subarachnoid

Since the arachnoid is punctured, CSF will be present. This may occur following removal of the loss of resistance syringe, where CSF will be seen as a continuous flow of clear fluid from the Tuohy needle, or after catheter insertion or migration, where clear fluid will be continuously aspirated. Symptoms will be

that of a spinal anaesthetic and will therefore depend upon the volume and concentration of anaesthetic injected. A rapid onset of motor block (e.g., after a test dose) should alert the anaesthetist to a subarachnoid block. See the following chapter for management of an accidental dural puncture.

Intravascular
This will usually be identified at the time of catheter insertion as the presence of blood in the catheter which can continuously be aspirated. If anaesthetic is injected symptoms will be related to the dose given and will be in accordance with local anaesthetic toxicity.

Unilateral Block
This can occur as a result of epidural placement in the lateral aspect of the epidural space or passage or migration through the intervertebral foramina.

Insufficient Activation
An appropriate dose of local anaesthetic is required to achieve analgesia via an epidural catheter. The dose required will be dependent on several factors including height, weight and stage of labour. Typically 10 – 15 mL of bag mix is used as a bolus following catheter insertion, after a test dose is given. Failure to give sufficient anaesthetic to establish the block will result in inadequate analgesia.

Catheter Migration
Epidural catheters can change their location after insertion as a result of patient movement and may become located in any anatomical structure in proximity to them (see tables above). The anaesthetist should be alert to this possibility, especially when giving bolus or top-up doses.

Abnormal Anatomy
A patient who has abnormal anatomy of their back may experience an insufficient block depending on the abnormality present. This should be a differential diagnosis particularly when the response to local anaesthetic is difficult to explain by other more common reasons.

Sacral Sparing
The S2-S4 nerve roots innervate the vagina and perineum and are therefore responsible for pain sensation in the second stage in labour. They are large, covered with thick dura mater and lie further from the tip of an epidural catheter than the T10 – L1 nerve roots. This can be a reason for failure of an epidural in the second stage of labour.

Precipitant Labour
If labour is precipitant there may be insufficient time to establish a functioning epidural.

Patient Expectations
Patients should be made fully aware of the degree to which analgesia will be provided by an epidural, and that some symptoms can be more difficult to ameliorate.

Management of Insertion and Placement Difficulties
Inability to Thread the Catheter
This should alert the anaesthetist to the catheter not being in the epidural space. Retraction of the catheter through the Tuohy needle whilst the needle is still in the back is not recommended due to the possibility of shearing off the catheter.

Blood in the Catheter
If blood is not able to be aspirated try flushing the catheter carefully; if no blood is present on repeat aspiration the catheter can be treated as normal. If blood is able to be aspirated, try withdrawing the catheter by 1 cm. If blood can still be aspirated then remove the catheter completely. If in doubt, take it out.

Paraesthesia

Some *transient* paraesthesia following insertion of the catheter is common. Persistent paraesthesia mandates removal of the catheter. Always document paraesthesia regardless of transience or permanence, including its location.

Accidental Dural Puncture

The epidural should be re-sited in an adjacent interspace. Caution must be exercised when the first dose of local anaesthetic is given through the catheter as there is a significant risk of intrathecal diffusion via the dural puncture site. If a normal response is obtained following this dose the epidural should then be conducted as per normal.

Management of Ineffective Epidurals

Assessment

Assess the patient's symptoms carefully, e.g., no block, partial or unilateral block, location and nature of pain (abdominal versus perineal, pressure versus discomfort etc).
Enquire as to whether the epidural has worked at any point. Assess for any improvement following bolus doses. Consider checking the catheter location at the back and compare with documentation of the catheter insertion. Look for evidence of fluid leakage at the insertion point. Assess motor and sensory block height. Enquire as to progress of labour and fetal position.
Consider a full bladder and uterine rupture as causes of breakthrough pain.

No block; low block

Give 10 - 15mL of standard mix with mother in normal position. Recheck at 20 minutes. Following this, if the block is still inadequate consider re-siting epidural at different interspace. If there was improvement after standard mix top up but she is still sore then it is appropriate to try to achieve denser block with top up of 7 – 10mL 0.25% bupivacaine with or without addition of fentanyl (50 – 100 µg).

Poor sacral coverage

Suspect this if mother is describing groin/vaginal/rectal/perineal pain. Check lateral aspect of feet for S1/S2 coverage. Give 10 - 15 mL of standard mix with mother sitting upright. Recheck block at 20 minutes. Consider addition of fentanyl (50 – 100 µg) with this bolus.
If sacral coverage remains inadequate consider re-site at different (preferably lower) interspace. If sacral coverage appears adequate on block testing with ethyl chloride but she is still sore then follow back pain/rectal pressure section.

Unilateral block

Pull catheter back by 1 – 2 cm (always leaving at least 3 cm in the epidural space, 4 cm if obese). Put patient into lateral position, sore side down and give 10 - 15 mL of standard mix solution. Recheck block in 20 minutes.
If still sore on affected side (pull back catheter if not already done so) then give 7 – 10 mL 0.25% bupivacaine with sore side down (consider adding fentanyl 50 – 100 µg with this bolus). Recheck block at 20 minutes. If block still inadequate, will need re-site.

Missed segment

Is it a true missed segment? This is frequently diagnosed when unilateral block is present. Manage initially as per unilateral block. Initially with bag mix +/- pulling catheter back (see unilateral block).
If a true missed segment, then give 7 – 10 mL 0.25% bupivacaine with sore side down (consider adding fentanyl 50 – 100 µg). Recheck block at 20 minutes. If still inadequate, consider re-site.
Remember the apt adage: If in doubt, take it out.

Back pain/rectal pressure

This is more likely in later stages (8 – 10 cm dilatation) with descent of the fetal head. Particularly common in OP presentations. If block height and sacral coverage appears adequate consider use of an adjuvant

(fentanyl) with 10 - 15 mL of standard mix solution with the mother sitting up.
If this is insufficient and block height appears adequate then denser block may help. Consider 7 – 10 mL 0.25% bupivacaine. Explain that complete relief of sensation of pressure may not be achievable. If analgesia still inadequate consider re-site.

Standard mix or 0.25% Levobupivacaine?
In most cases of breakthrough pain, successful analgesia can be achieved with larger volumes of standard mix to achieve better spread. Adjuncts like fentanyl are very helpful. Avoiding 0.25% bupivacaine when troubleshooting may reduce motor block and improve mode of delivery. However, 0.25% bupivacaine is very useful to help establish efficacy or otherwise. Inadequate analgesia following a top up with 0.25% bupivacaine suggests the epidural is not in the right place and should be re-sited.

Adjuvant Drugs
Remember the catheter dead space (1 mL) which is rather significant if giving undiluted fentanyl. Follow on with bag mix bolus.

Fentanyl
Epidurally administered fentanyl provides rapid onset of excellent analgesia. It can be especially useful in the case of sacral sparing. Typical dose 50 – 100 µg.

Re-siting
All anaesthetists should be humble with regards to epidurals and persistent attempts to manage an ineffective epidural should be avoided. It is beneficial to attend promptly to troubleshooting requests and promptly (20 minutes) reassess after intervention (e.g., bolus). If appropriate management techniques as noted above have failed to provide suitable analgesia, then the anaesthetist should offer re-site. Depending on the circumstances there may need to be a discussion regards risk/benefit of re-siting. If uncertain, seek senior help.

References
Hermanides, J., Hollmann, M.W., Stevens M.F., Lirk, P. 2012. Failed epidural: causes and management. *British Journal of Anaesthesia*, 109(2): 144-154.
Kingsely, C., McGlennan, A., Brown, J.P., Abir, G. 2017. The labour epidural: troubleshooting. *Anaesthesia Tutorial of the Week, World Federation of Societies of Anaesthesiologists*. Available at ResearchGate.
West Suffolk NHS Foundation Trust. 2017. Regional analgesia in labour. Obstetric Anaesthetists Association example guidelines. Available at: https://www.oaa-anaes.ac.uk/assets/_managed/cms/files/West%20Suffolk%20-%20RegionalAnalgesia,September2017.pdf.
Greater Glasgow and Clyde NHS Trust. Troubleshooting labour epidurals. Obstetric Anaesthesia association example guidelines. Available at: https://www.oaaanaes.ac.uk/assets/_managed/cms/files/Clinical%20Guidelines/Glasgow%20Epidural%20troubleshooting.pdf.

🛠 A poorly functioning labour epidural should not be topped-up for epidural anaesthesia. Rather, it should be removed and a spinal or CSE should be performed instead, if appropriate.

Treatment of Post-Dural Puncture Headache

The development of post-dural puncture headache (PDPH), or spinal headache, is characterised by severe, disabling fronto-occipital pain with radiation to neck and shoulders. There may be neck stiffness. The pain may be completely relieved by lying supine. Although onset is commonly on the first or second day after dural puncture, it may occur on the same day and up to five days later. PDPH remits spontaneously within two weeks, but is associated with a small but significant increase in subdural haematoma, so is not a completely benign condition.

Management

Inform the consultant on call.
Exclude other causes of severe headache.
Document the PDPH by marking the appropriate boxes on the audit form (Headache Mild or Severe; Headache Spinal; Dural Tap).
Complete the PDPH form
Unless there is a contraindication, prophylactic dalteparin should be prescribed for 22:00 for the inpatient stay.
Discuss the problem and its management with the patient. Give her the PDPH information leaflet to read.
There are basically two options depending on how disabling the headache is:

Conservative management

Encourage oral fluids and simple analgesia: paracetamol, ibuprofen or diclofenac, dihydrocodeine and/or tramadol (prescribe regular analgesics and an as required analgesic) unless contraindicated.
Recording of 4-hourly observations must be initiated.

Pharmacological management

Caffeine

There is limited evidence to support the use of caffeine. If used, the dose should be less than 300 mg, with a total dose not exceeding 900 mg in 24 hours. For breastfeeding women, especially those with low birth weight or premature babies, a total 24-hour dose of 200 mg should be considered.
It is difficult to provide accurate values for the caffeine content of drinks. As an approximate guide, a mug of coffee will contain 65 – 160 mg of caffeine; an espresso will contain 45 – 100 mg. Coca cola® and Coke Zero® contain 32 mg (in one 330 mL can); Diet Coke® contains 42 mg per can. Tea contains 25 – 50 mg.

Other medication

There is insufficient evidence to support the use of other agents including theophyllines, steroids, triptans and gabapentinoids.

Epidural blood patch

An epidural blood patch (EBP) should be considered when conservative management is ineffective and the woman experiences symptoms which impair functions of daily living, including care of her baby. They must be discussed with the consultant on call. Informed consent must be obtained.
Current evidence suggests that one third of patients receiving an EBP will experience complete and permanent relief of symptoms. Complete or partial relief will occur in 50 – 80% of patients. In cases of partial or no relief, a second EBP can be performed provided that other causes of symptoms have been excluded.
The patient should be apyrexial – check the white blood count if indicated. Coagulation should be normal – if on low molecular weight heparin, appropriate timing is essential (i.e., at least 12 hours following a

prophylactic dose or 24 hours following a treatment dose).
Although it has been suggested that efficacy may be improved by delaying epidural blood patch until 48 hours after dural puncture, this delay is not recommended as delay may increase the risk of serious sequelae including subdural haematoma. The timing for performing the blood patch is at the consultant anaesthetist's discretion.

Epidural blood patch technique

A Tuohy needle is sited in the epidural space overlying the puncture site or one interspace below using a standard epidural technique.
At least 20 mL of autologous blood are taken from the patient under sterile conditions.
20 mL of this blood are injected via the Tuohy needle into the epidural space. If the patient complains of discomfort in her back or down her legs the injection should be stopped.
The patient lies flat for at least 2 hours and then mobilises gradually. She should be advised to avoid vigorous activity and straining for several days.
Details about the procedure should be completed on PDPH Audit form. The standard letter to the patient's GP should be completed. The patient should be given a Post Dural Puncture Information Leaflet (available in Jenna's office) and contact information for medical advice.
Follow-up is important and should as a minimum be done telephonically after one week and one month.

References

Paech, M.J. 2005. Epidural blood patch – myths and legends. *Canadian Journal of Anaesthesia*, 52(s1): R1-5.
Sprigge, J.S., Harper, S.J. 2008. Accidental dural puncture and post dural puncture headache in obstetric anaesthesia: presentation and management: A 23-year survey in a district general hospital. *Anaesthesia*, 63: 36-43.
Van de Velde, M., Schepers, R., Berends, N. *et al*. 2008. Ten years experience with ADP and PDPH in a tertiary obstetric anaesthesia department. *International Journal of Obstetric Anesthesia*, 17(4): 329-35.
Paech, M.J. 2012. Iatrogenic headaches: giving everyone a sore head. *International Journal of Obstetric Anesthesia*, 21(1): 1-3.
Scavone, B.M. 2015. Timing of epidural blood patch: clearing up the confusion. *Anaesthesia*, 70: 119-21.
Obstetric Anaesthetists' Association. 2018. *Treatment of obstetric post-dural puncture headache*. London: Obstetric Anaesthetists' Association.

High Regional or Total Spinal Block

The incidence is 1 in 27,000 obstetric epidurals. Strictly, total spinal is the subarachnoid injection of a large

(epidural) dose of local anaesthetic, resulting in severe hypotension, profound bradycardia, respiratory arrest and loss of consciousness. However, a similar result can follow epidural or subdural injection of local anaesthetic: high regional block is defined as an excessively high block requiring tracheal intubation. Although the features (weakness of upper arms, difficult breathing, slurred speech, sedation and high level of numbness) are typically rapidly ascending, these may develop late and insidiously.

Management

Call immediately for a second anaesthetist.

Equipment immediately required

Cardiac arrest trolley with laryngoscope, endotracheal tubes and suction pump.

Management technique

- Position patient on her left side (or other method of uterine displacement). The patient must be positioned to eliminate aortocaval compression at all times.
- Oxygen should be administered with a mask and bag as necessary.
- 500 - 1000 mL of Hartmann's solution rapidly intravenously.
- Ephedrine 15 mg intravenously. Repeat as necessary. Consider phenylephrine, adrenaline and atropine administration. Maintain the blood pressure with ephedrine or a phenylephrine infusion plus atropine and an intravenous infusion as required.
- Move the patient to the operating theatre for full monitoring including blood pressure, SpO2, EtCO2 and ECG.
- If breathing is inadequate, perform rapid sequence induction with cricoid pressure. The patient should be intubated and kept ventilated and anaesthetised.
- If intubation is difficult or fails, do not wait for the return of spontaneous ventilation – it may not occur. Proceed with emergency airway management and IPPV (see failed intubation drill).
- In the event of fetal distress, caesarean section is advised.
- The patient and her relative(s) should be reassured; allocate a team member to escort the partner/relative out of the room as soon as possible.
- Consider alternative diagnoses, e.g., subarachnoid haemorrhage.

Post-event

A post-delivery visit is vital to explain what has happened and why. Further follow up may be needed to deal with psychological issues.
An adverse incident report must be filed.

References

Yentis, S.M. 2001. High regional block – the failed intubation of the new millennium. *International Journal of Obstetric Anesthesia,* 10: 159-61.